ECEN 5113

HW: 7

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Procedure:

To complete this lab, Verilog files were first created for both the array multiplier and IEEE multiplier. These were synthesized in Vivado to ensure the behavior of the code worked properly. Once this was done, a netlist was created for both files using the Cadence Innovus Implementation System. Innovus was then used to create the layout of the circuits. Connectivity, DRC, and timing checks were done to verify that the layouts were properly created.

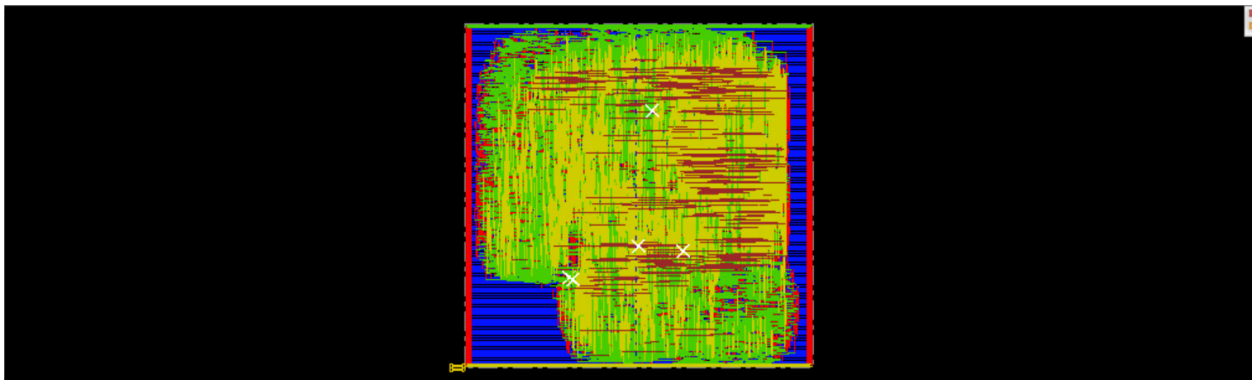


Figure 1: Layout of 64x64 Array Multiplier

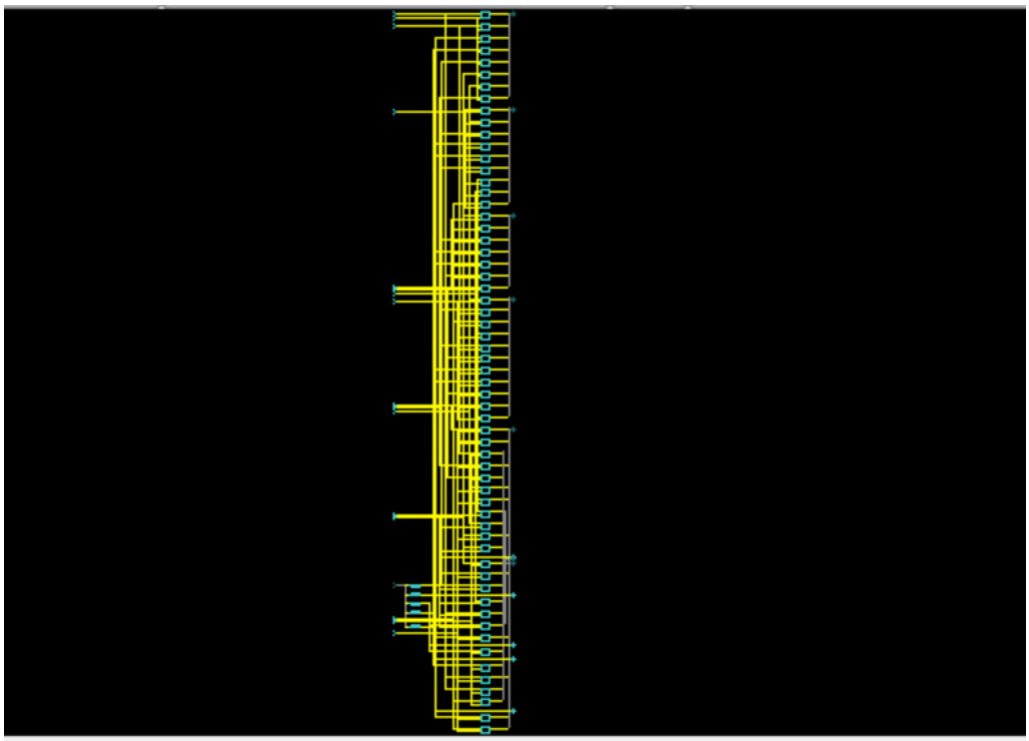


Figure 2: Schematic of 64x64 Array Multiplier

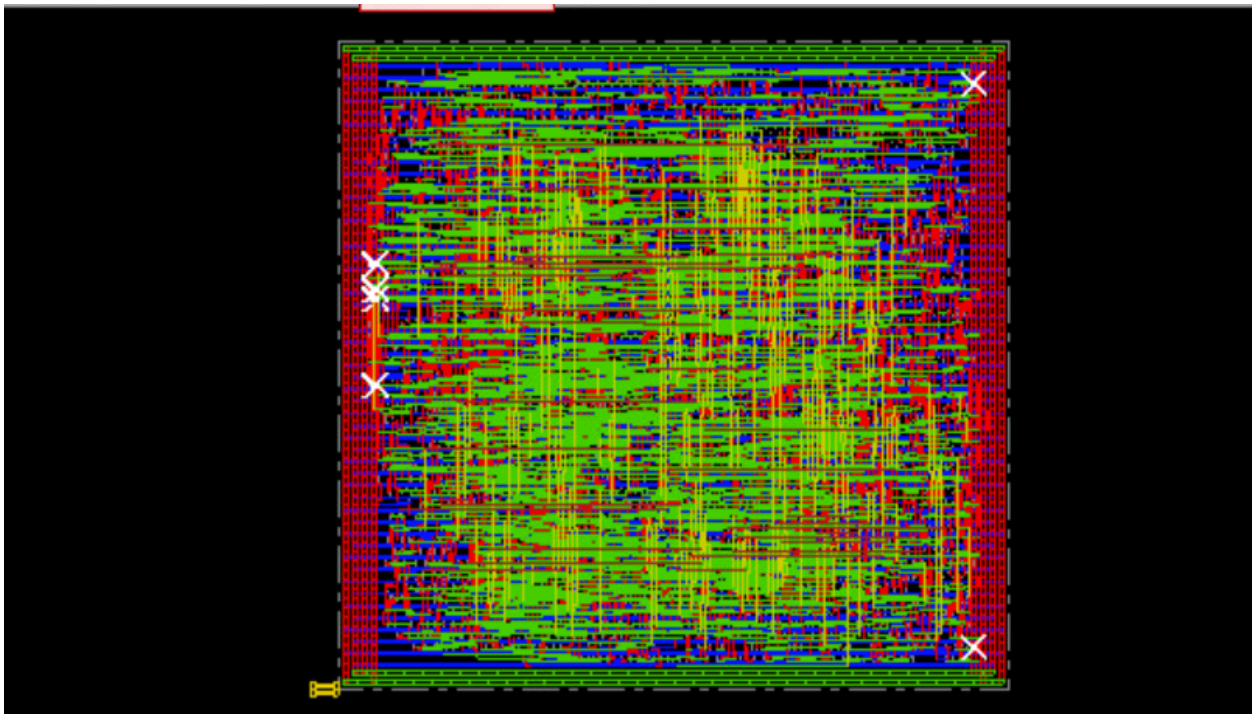


Figure 3: Layout of IEEE Multiplier

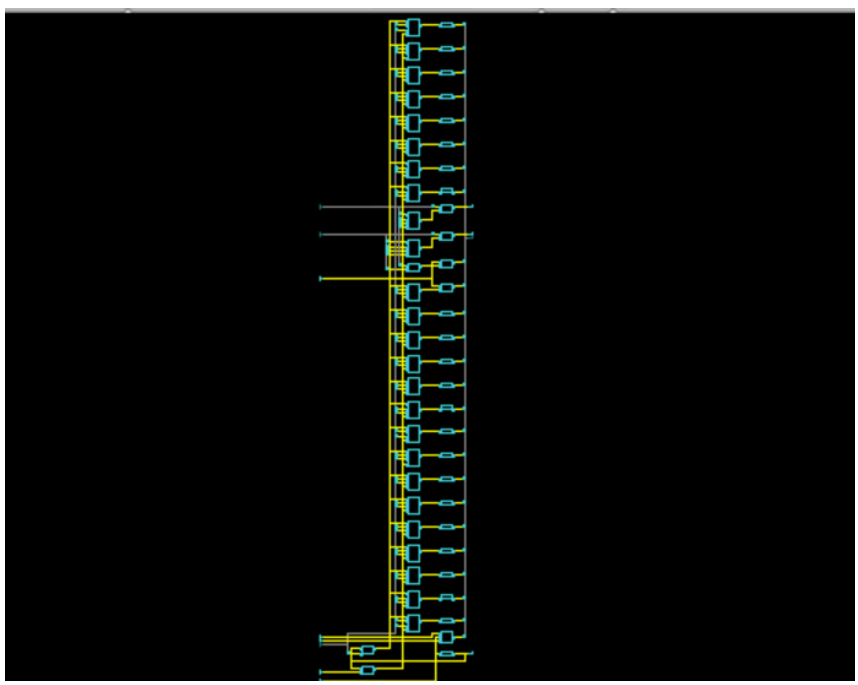


Figure 4: Schematic of IEEE Multiplier

Observations;

There were two violations found when running a DRC check on the IEEE multiplier, but it was few enough were it could warrant ignoring. The violation report showed that there was a short

on Metal 2 somewhere on the circuit. The timing report for the IEEE multiplier showed that the design is very slow. This is most likely because the multiplication logic is happening in a ripple-carry way. To fix this, the design could be pipelined. The timing report for the 64x64 array multiplier also showed to be slow. There is a fanout of 64 at some point, which causes a big delay. Again, pipelining the design would help decrease the overall delay.

Results:

64x64 Array Multiplier

The figures below show the output for the 64x64 array multiplier. The first ones show the output from synthesizing the Verilog code in Vivado. The first case shows when 3 times 5 is multiplied and gets the correct output of 15. The second case shows that when multiplying a random integer by 0, the output is 0. The third case shows multiplication with a big number. Test case four shows that when multiplying by the two maximum values, the output can hold the maximum result.

```
# }
# run 1000ns
time=10000 | a=3 | b=5 | o=15
time=20000 | a=0 | b=78944 | o=0
time=30000 | a=4294967295 | b=4 | o=17179869180
time=40000 | a=18446744073709551615 | b=18446744073709551615 | o=340282366920938463426481119284349108225
```

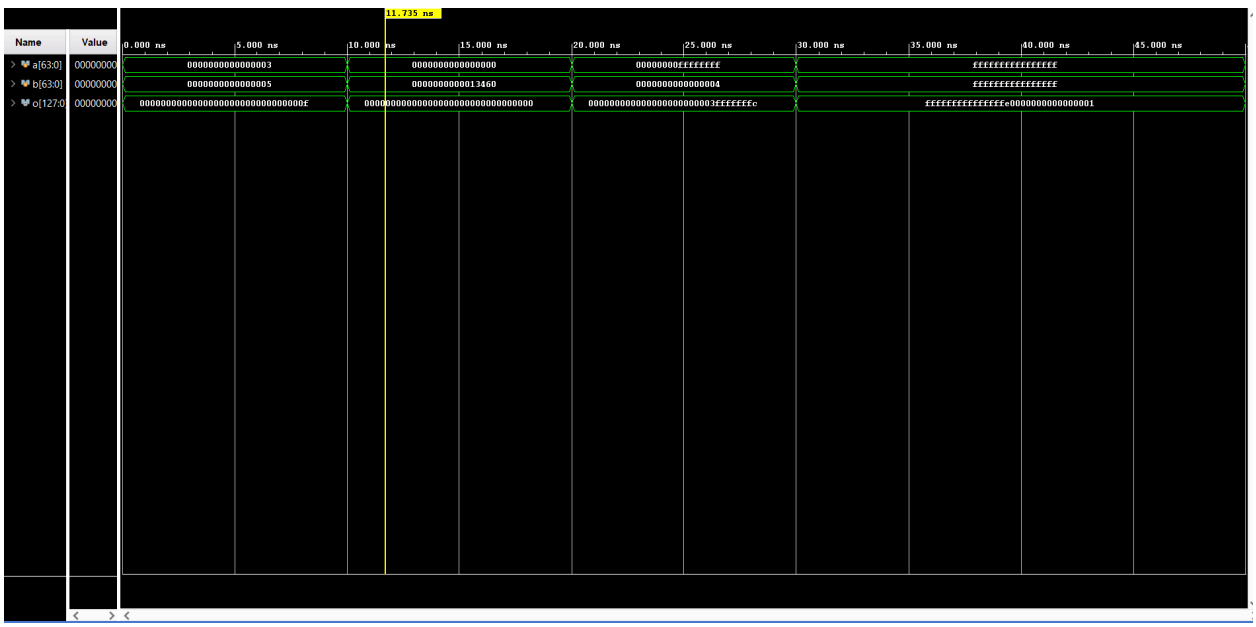


Figure 5: Vivado Output of 64x64 Array Multiplier

Pre-CTS Timing

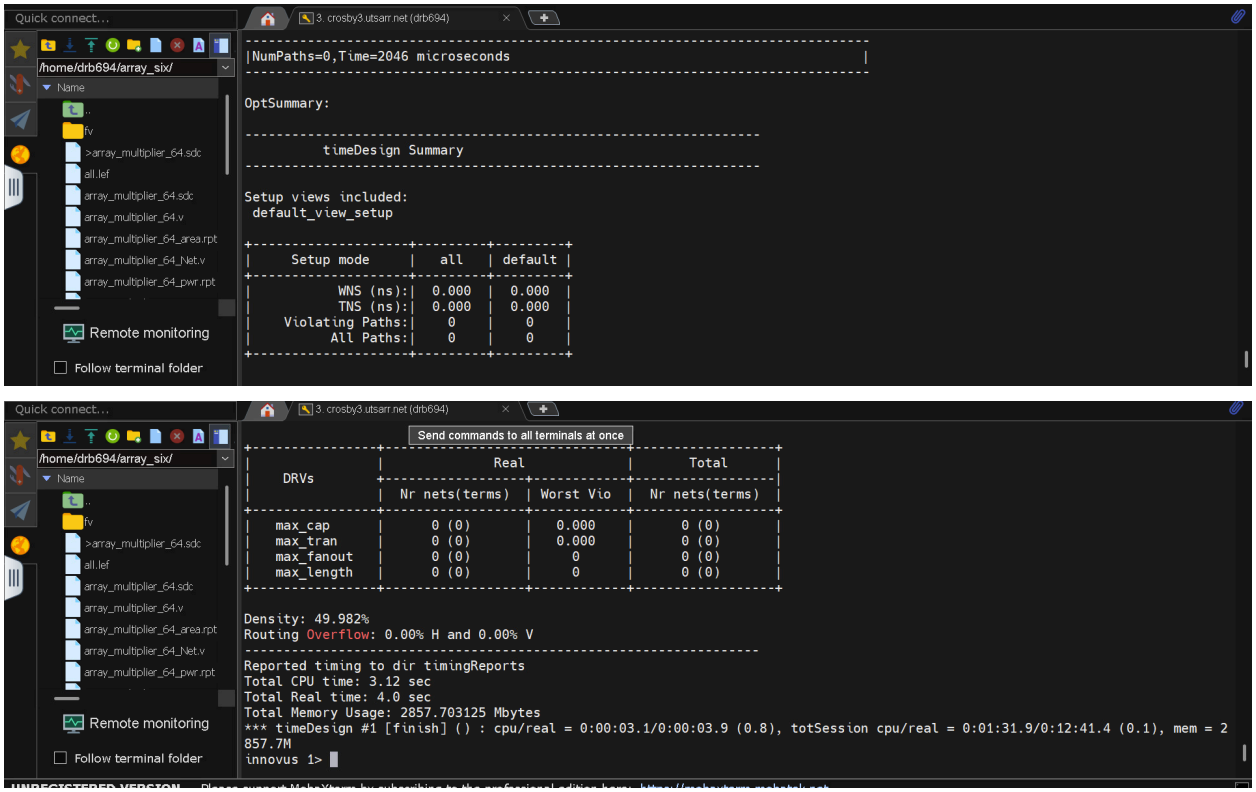
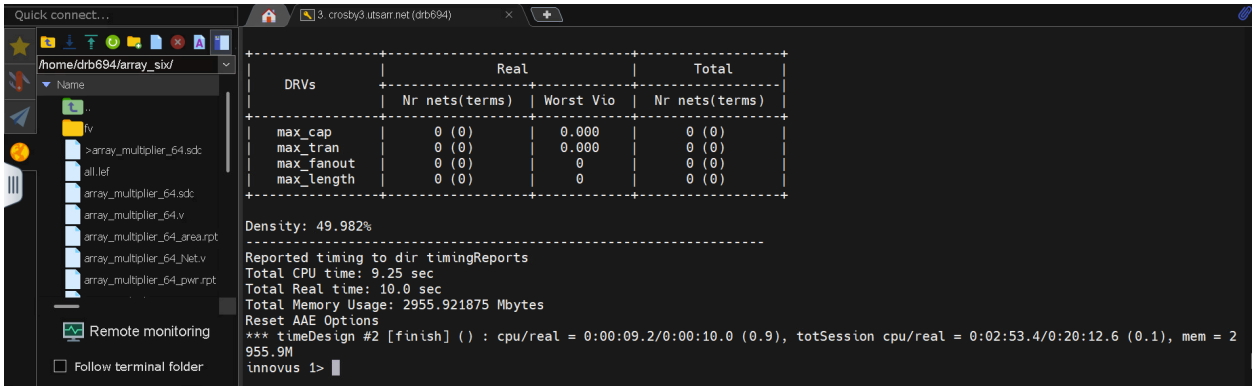


Figure 6: Timing of 64x64 Array Multiplier

Post Route Timing



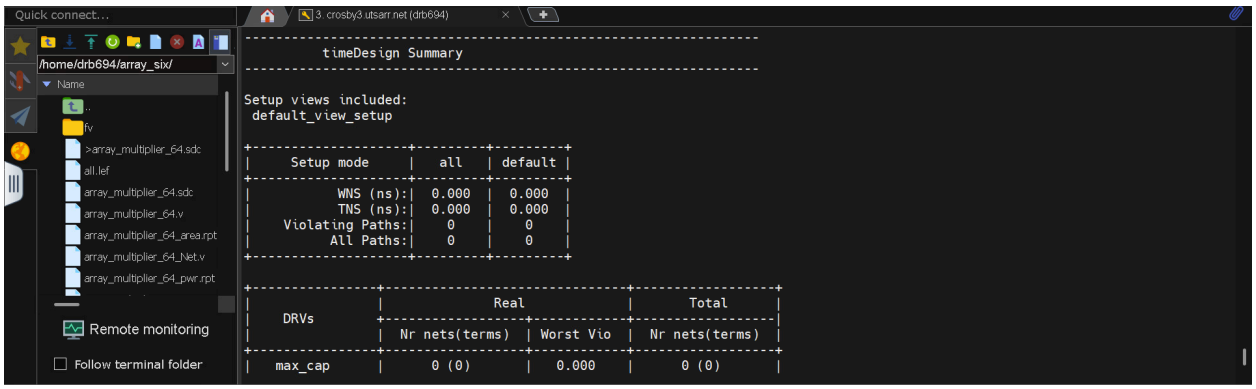
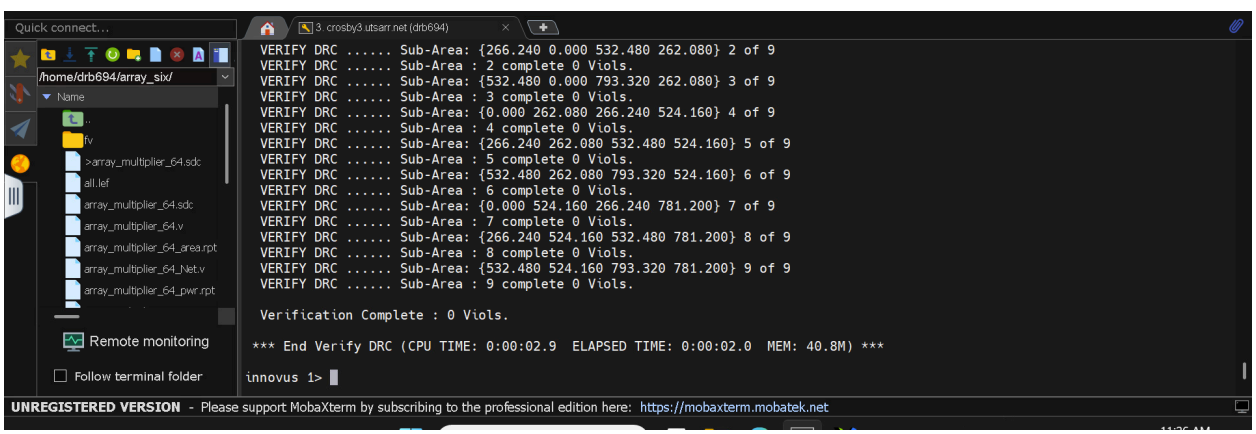


Figure 7: Post Route Timing of 64x64 Array Multiplier

DRC



```
File    Edit    View

#############################################################
#  Generated by:      Cadence Innovus 25.11-s102_1
#  OS:                Linux x86_64(Host ID crosby3.utsarr.net)
#  Generated on:      Fri Nov 14 11:26:11 2025
#  Design:            array_multiplier_64
#  Command:           verify_drc
#############################################################
#set_verify_drc_mode -check_ndr_spacing auto -check_only default -check_same_via_cell true -disable_rules "" -exclude_pg_net false -ignore_cell_blockage
false -ignore_trial_route false -limit 1000 -report array_multiplier_64.drc.rpt -use_min_spacing_on_block_obs auto

No DRC violations were found

Ln 1, Col 1    663 characters    100%    Unix (LF)    UTF-8
```

Figure 8: DRC Check of 64x64 Array Multiplier

Connectivity Check

```
Quick connect...    3. crosby3.utsarr.net (drb694)
/home/drb694/array_six/
Name
..
fv
>array_multiplier_64.sdc
all.lef
array_multiplier_64.sdc
array_multiplier_64.v
array_multiplier_64_area.rpt
array_multiplier_64_Net.v
array_multiplier_64_pwr.rpt
Remote monitoring
Follow terminal folder

**WARN: (IMPVFC-97):    IO pin a[45] of net a[45] has not been assigned. Please make sure it is assigned and rerun verifyConne
ctivity.
**WARN: (IMPVFC-97):    IO pin a[44] of net a[44] has not been assigned. Please make sure it is assigned and rerun verifyConne
ctivity.
**WARN: (EMS-27):       Message (IMPVFC-97) has exceeded the current message display limit of 20.
To increase the message display limit, refer to the product command reference manual.
**** 11:27:28 **** Processed 5000 nets.
**** 11:27:28 **** Processed 10000 nets.
**** 11:27:28 **** Processed 15000 nets.

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Fri Nov 14 11:27:29 2025
Time Elapsed: 0:00:01.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.6  MEM: 8.305M)

innovus 1>

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
```

```
File    Edit    View

#############################################################
#  Generated by:      Cadence Innovus 25.11-s102_1
#  OS:                Linux x86_64(Host ID crosby3.utsarr.net)
#  Generated on:      Fri Nov 14 11:27:28 2025
#  Design:            array_multiplier_64
#  Command:           verifyConnectivity -type all -geomConnect -error 1000 -warning 50
#############################################################
Verify Connectivity Report is created on Fri Nov 14 11:27:28 2025

Begin Summary
    Found no problems or warnings.
End Summary
```

Figure 9: Connectivity Check of 64x64 Array Multiplier

Check Placement

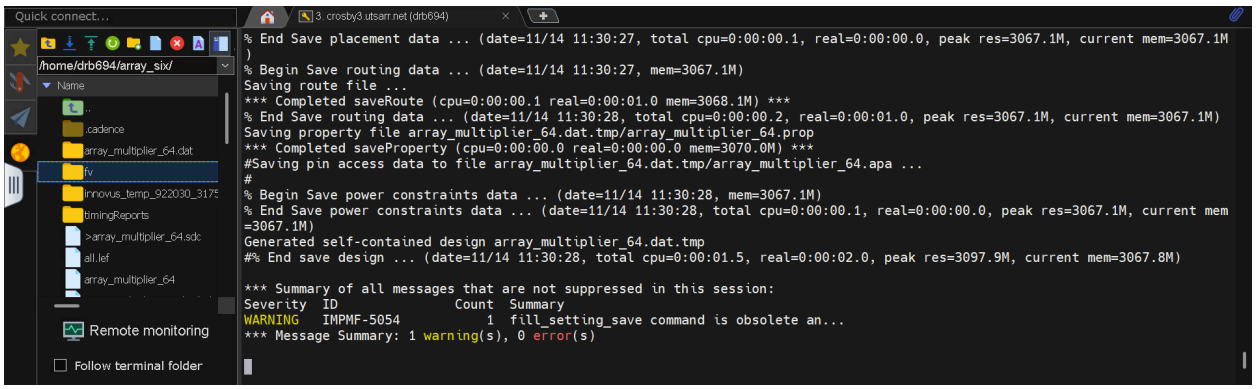
```
#################################################################
#  Generated by:      Cadence Innovus 25.11-s102_1
#  OS:                Linux x86_64(Host ID crosby3.utsarr.net)
#  Generated on:      Fri Nov 14 11:14:01 2025
#  Design:            array_multiplier_64
#  Command:           checkPlace array_multiplier_64.checkPlace
#################################################################

## No violations found ##

## Summary:
#######################################################
## Number of Placed Instances = 7222
## Number of Unplaced Instances = 0
## Placement Density:49.98%(293734/587675)
## Placement Density (including fixed std cells):49.98%(293734/587675)
```

Figure 10: Check Placement of 64x64 Array Multiplier

GDS2



```
    Via Instances                    0

Metal FillDRCs                       0

    Via Instances                    0

Text                                 0


Blockages                            0


Custom Text                          0


Custom Box                           0

Trim Metal                           0

######Streamout is finished!
innovus 1>
```

```
Stream Out Layer Mapping Information:
GDS Layer Number          GDS Layer Name
----------------------------------------


Stream Out Information Processed for GDS version 3:
Units: 2000 DBU

Object                              Count
----------------------------------------
Instances                            7222

Ports/Pins                              0

Nets                                    0

    Via Instances                       0

Special Nets                            0

    Via Instances                       0
```

Figure 11: GDS2 of 64x64 Array Multiplier

Area Report

```
array_multiplier_64_area.rpt
File   Edit   View

            Instance             Cells  Cell Area  Net Area  Total Area  Wireload
-----------------------------------------------------------------------------------
array_multiplier_64               7222     293734         0      293734    <none> (D)
  csa_tree_add_81_43_I64_groupi   2998     252008         0      252008    <none> (D)

 (D) = wireload is default in technology library
=============================================================
  Generated by:           Encounter(R) RTL Compiler v14.10-s022_1
  Generated on:           Nov 14 2025  11:01:33 am
  Module:                 array_multiplier_64
  Technology library:     tsmc18 1.0
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
=============================================================

            Instance             Cells  Cell Area  Net Area  Total Area  Wireload
-----------------------------------------------------------------------------------
array_multiplier_64               7222     293734         0      293734    <none> (D)
  csa_tree_add_81_43_I64_groupi   2998     252008         0      252008    <none> (D)

 (D) = wireload is default in technology library

Ln 1, Col 1   1,688 characters   100%   Unix (LF)   UTF-8
```

Figure 12: Area Report of 64x64 Array Multiplier

```
                                   Leakage     Dynamic       Total
          Instance          Cells Power(nW)   Power(nW)    Power(nW)
-----------------------------------------------------------------------
array_multiplier_64          7222 13720.493 80021100.589 80034821.082
  csa_tree_a..43_I64_groupi  2998 13171.355 77565070.388 77578241.743

============================================================
  Generated by:           Encounter(R) RTL Compiler v14.10-s022_1
  Generated on:           Nov 14 2025  11:01:33 am
  Module:                 array_multiplier_64
  Technology library:     tsmc18 1.0
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

                                   Leakage     Dynamic       Total
          Instance          Cells Power(nW)   Power(nW)    Power(nW)
-----------------------------------------------------------------------
array_multiplier_64          7222 13720.493 80021100.589 80034821.082
  csa_tree_a..43_I64_groupi  2998 13171.355 77565070.388 77578241.743
```

Figure 13: Power Report of 64x64 Array Multiplier

```
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

    Pin          Type     Fanout  Load Slew Delay Arrival
                                  (fF) (ps)  (ps)   (ps)
-------------------------------------------------------------
a[0]            in port        1   3.5    0    +0       0 F
g32874/A                                         +0       0
g32874/Y        INVX1         64 204.8 2511 +1384    1384 R
g29763/A                                         +0    1384
g29763/Y        NOR2XL         1   6.8  406  +153    1537 F
csa_tree_add_81_43_I64_groupi/in_59[4]
  g203496/B                                      +0    1537
  g203496/CO    ADDFX2         1   4.4  140  +595    2132 F
  g204749/D                                      +0    2132
  g204749/S     CMPR42X1       1   6.9  128  +786    2917 F
  g3147/A                                        +0    2917
  g3147/CO      ADDFX2         1   2.7  134  +483    3400 F
  g205093/ICI                                    +0    3400
  g205093/CO    CMPR42X1       1   2.7  100  +310    3710 F
  g205092/ICI                                    +0    3710
  g205092/CO    CMPR42X1       1   6.2  124  +325    4035 F
```

Figure 14: Timing Report of 64x64 Array Multiplier

IEEE Multiplier

The figures below show the output for the IEEE multiplier. The first two figures show the output when the Verilog code was synthesized in Vivado. The first case tests 2 times 3 and gets the correct output 6. The second case test when 0 is multiplied by infinity and shows that the output is NaN. The third case shows that when infinity is multiplied by -1, the output is negative infinity. The fourth case shows that when 1.5 is multiplied by 1, the output is 1.5. The fifth case shows that when -2 is multiplied by 3, the output is -6. The sixth case shows that when NaN is multiplied by NaN, the output is NaN.

```
# run 1000ns
time=10000 | a=40000000 | b=40400000 | o=40c00000
time=20000 | a=0 | b=7f800000 | o=7fc00000
time=30000 | a=7f800000 | b=bf800000 | o=ff800000
time=40000 | a=3fc00000 | b=3f800000 | o=3fc00000
time=50000 | a=c0000000 | b=40400000 | o=c0c00000
time=60000 | a=7fc00000 | b=3f800000 | o=7fc00000
```

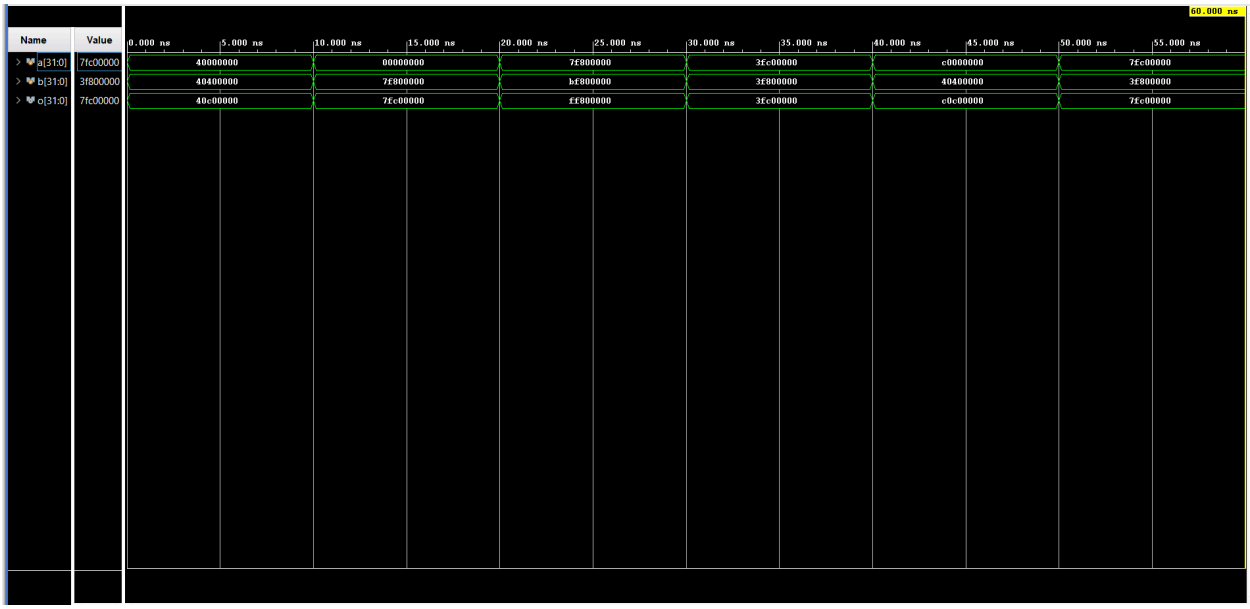


Figure 15: Vivado Output of IEEE Multiplier

Pre-CTS Timing Report

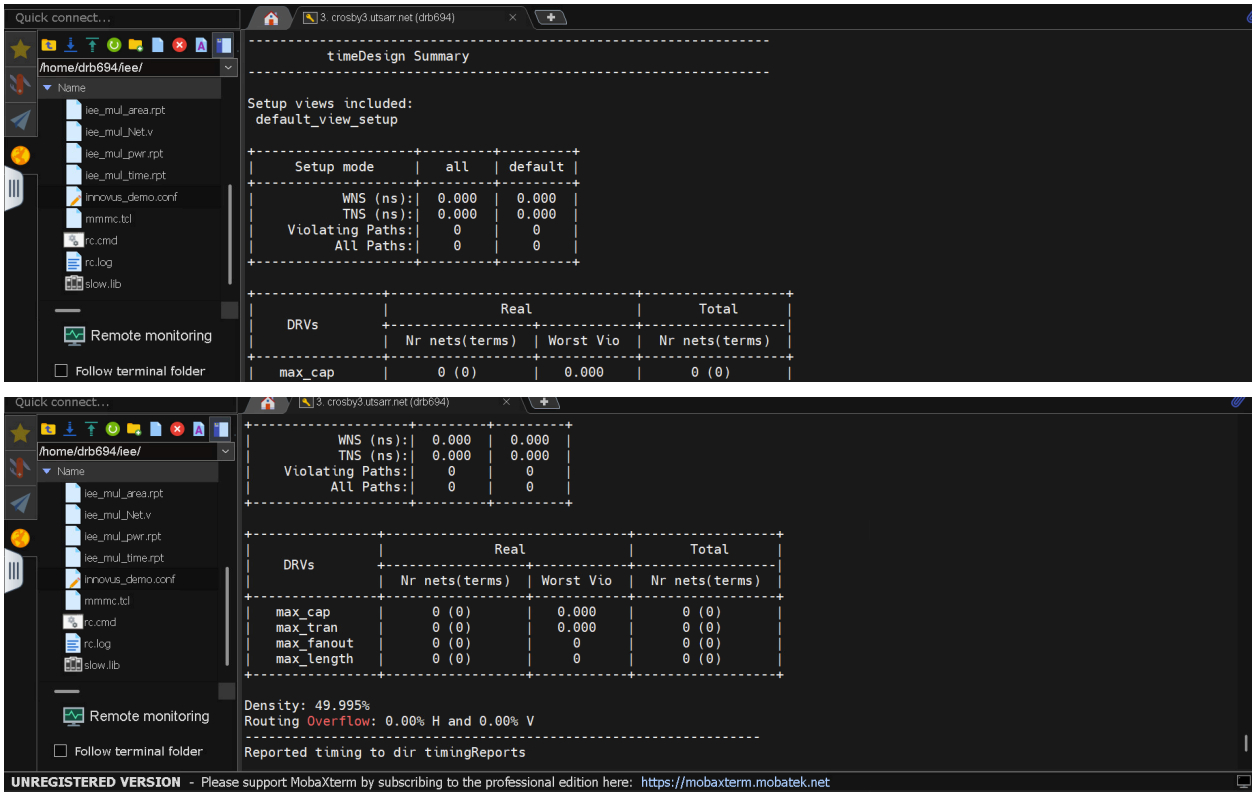


Figure 16: Pre-CTS Timing Report of IEEE Multiplier

Post-Route Timing Report

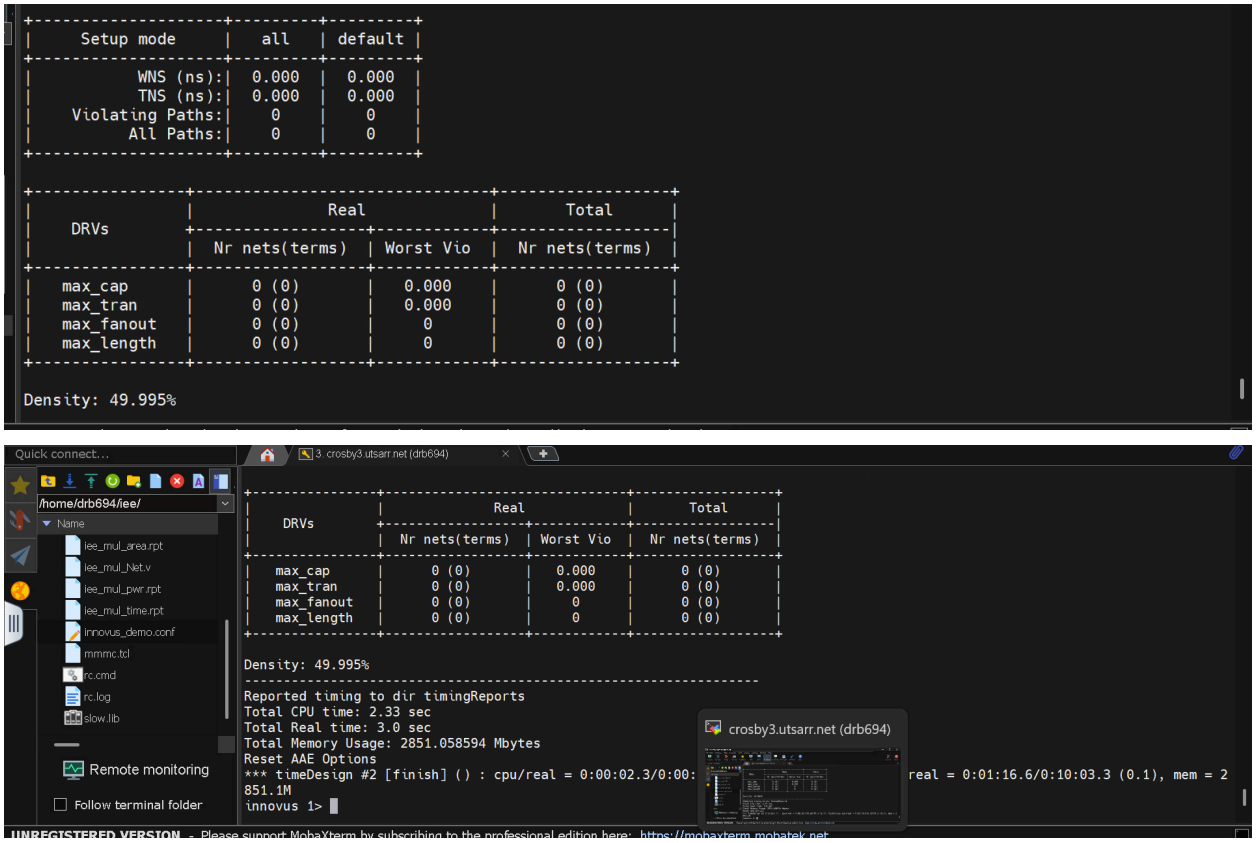
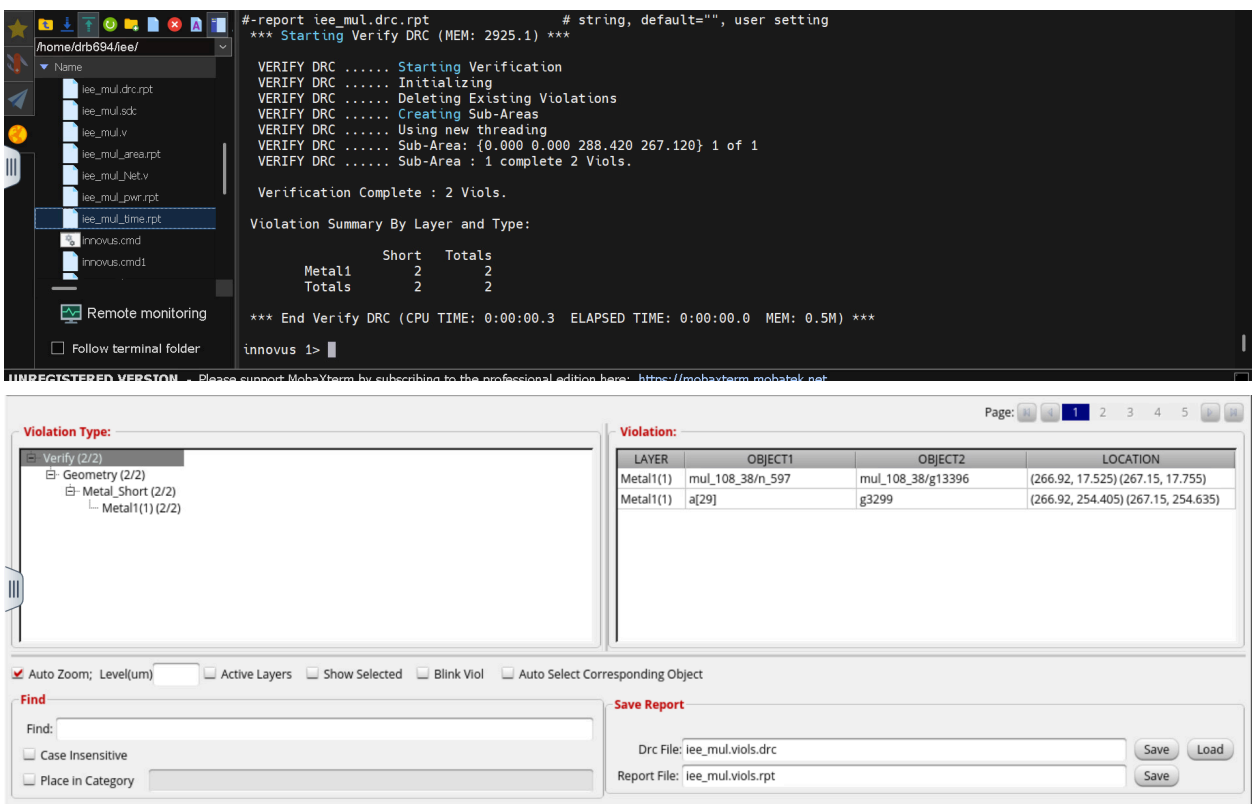


Figure 17: Post-Route Timing Report of IEEE Multiplier

DRC



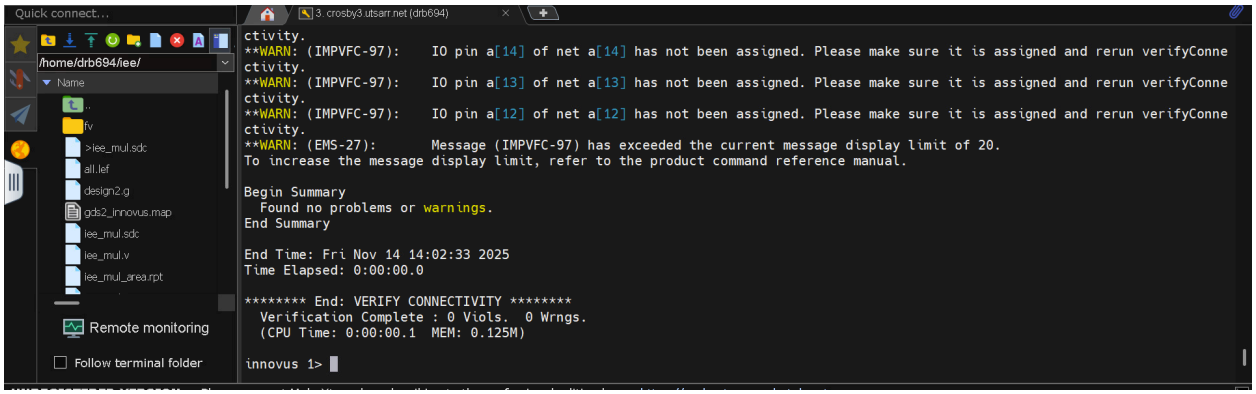
```
#############################################################
#  Generated by:      Cadence Innovus 25.11-s102_1
#  OS:                Linux x86_64(Host ID crosby3.utsarr.net)
#  Generated on:      Fri Nov 14 14:09:10 2025
#  Design:            iee_mul
#  Command:           verify_drc
#############################################################
#set_verify_drc_mode -area {0 0 0 0} -check_ndr_spacing auto -check_only default -check_same_via_cell true -disable_rules "" -exclude_pg_net false -
ignore_cell_blockage false -ignore_trial_route false -limit 1000 -report iee_mul.drc.rpt -use_min_spacing_on_block_obs auto

SHORT: ( Metal Short ) Regular Wire of Net mul_108_38/n_597 & Blockage of Cell mul_108_38/g13396  ( Metal1 )
Bounds : ( 266.920, 17.525 ) ( 267.150, 17.755 )


SHORT: ( Metal Short ) Regular Wire of Net a[29] & Blockage of Cell g3299  ( Metal1 )
Bounds : ( 266.920, 254.405 ) ( 267.150, 254.635 )


  Total Violations : 2 Viols.
```

Connectivity




```
#############################################################
#  Generated by:      Cadence Innovus 25.11-s102_1
#  OS:                Linux x86_64(Host ID crosby3.utsarr.net)
#  Generated on:      Fri Nov 14 14:02:33 2025
#  Design:            iee_mul
#  Command:           verifyConnectivity -type all -geomConnect -error 1000 -warning 50
#############################################################
Verify Connectivity Report is created on Fri Nov 14 14:02:33 2025



Begin Summary
    Found no problems or warnings.
End Summary
```

Ln 1, Col 1 | 538 characters | 100% | Unix (LF) | UTF-8

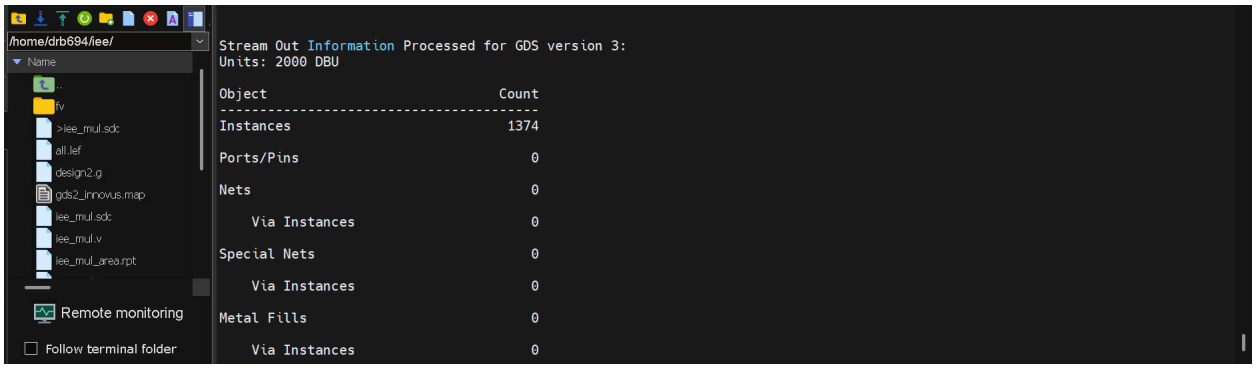
Check Placement

```
#################################################################
#  Generated by:      Cadence Innovus 25.11-s102_1
#  OS:                Linux x86_64(Host ID crosby3.utsarr.net)
#  Generated on:      Fri Nov 14 13:58:50 2025
#  Design:            iee_mul
#  Command:           checkPlace iee_mul.checkPlace
#################################################################

## No violations found ##

## Summary:
#########################################################
## Number of Placed Instances = 1374
## Number of Unplaced Instances = 0
## Placement Density:49.99%(32928/65862)
## Placement Density (including fixed std cells):49.99%(32928/65862)
```

GDS2



Power Report

```
                    Leakage    Dynamic      Total
  Instance   Cells Power(nW)  Power(nW)   Power(nW)
-----------------------------------------------------
iee_mul       1374  1223.911 5408152.852 5409376.763
  mul_108_38  1207  1120.244 5149853.546 5150973.790

============================================================
  Generated by:           Encounter(R) RTL Compiler v14.10-s022_1
  Generated on:           Nov 14 2025  01:54:40 pm
  Module:                 iee_mul
  Technology library:     tsmc18 1.0
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

                    Leakage    Dynamic      Total
  Instance   Cells Power(nW)  Power(nW)   Power(nW)
-----------------------------------------------------
iee_mul       1374  1223.911 5408152.852 5409376.763
  mul_108_38  1207  1120.244 5149853.546 5150973.790
```

Area Report

```
  Instance    Cells  Cell Area  Net Area  Total Area  Wireload
-----------------------------------------------------------------
iee_mul        1374      33008         0       33008    <none> (D)
  mul_108_38   1207      30124         0       30124    <none> (D)

 (D) = wireload is default in technology library
============================================================
  Generated by:           Encounter(R) RTL Compiler v14.10-s022_1
  Generated on:           Nov 14 2025  01:54:40 pm
  Module:                 iee_mul
  Technology library:     tsmc18 1.0
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

  Instance    Cells  Cell Area  Net Area  Total Area  Wireload
-----------------------------------------------------------------
iee_mul        1374      33008         0       33008    <none> (D)
  mul_108_38   1207      30124         0       30124    <none> (D)

 (D) = wireload is default in technology library
```

Timing Report

```
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

    Pin         Type      Fanout  Load Slew Delay Arrival
                                  (fF) (ps)  (ps)   (ps)
-------------------------------------------------------------
a[24]          in port         3  11.2    0    +0       0 F
g1526/D                                          +0       0
g1526/Y        OR4X1           1   4.3  128  +490     490 F
g1523/A                                          +0     490
g1523/Y        OR4X2          44 118.3  470  +594    1084 F
g1518/A                                          +0    1084
g1518/Y        AND2X1         21  74.8  513  +602    1686 F
mul_108_38/A[1]
  g13960/A                                       +0    1686
  g13960/Y     INVX1           9  29.4  413  +366    2051 R
  g13902/A                                       +0    2051
  g13902/Y     NOR2XL          1   6.2  138  +116    2167 F
  g13763/CI                                      +0    2167
  g13763/CO    ADDFX2          3   8.9  154  +357    2524 F
  g13667/B0                                      +0    2524
  g13667/Y     OAI21XL         1   3.0  256  +113    2637 R
  g13652/B0                                      +0    2637
```

Code:

64x64 Array Multiplier

```
////////////////////////////////////////////////////////////////////////////////


//64x64 array multiplier

//define module
module array_multiplier_64(
    input [63:0] a, //input
    input [63:0] b, //input
    output [127:0] o //output -- 64*2


    );


    //64^2 = 4096 - #of partial products

    //64 rows - 64 columns -- 64x64 matrix = 4096
    //eacn row = one partial product?
    wire [63:0] partial_product [63:0]; //partial product array

    //generate statement -- allows for conditional/iterative generation of hardware
    //builds multiple copies of hardware during synthesis

    //partial products
    //creates 4096 AND gates - generate one bit of the partial prodcut
    //partial_product[0][0] = a[0] & b[0]
    //partial product [0][1] = a[1] & b[0]
    //partial prod [63][63] = a[63] & b[63]


    genvar i, j; //generate variable

    generate
    //64 rows
```

Ln 1, Col 1 | 2,169 characters | Plain text | 100% | Unix (LF) | UTF-8

```
    generate
    //64 rows
        for(i=0; i<64; i=i+1) begin : partial_gen
            //64 columns
            for(j=0; j<64; j=j+1) begin : dig_gen
                assign partial_product[i][j] = a[j] & b[i];
            end
        end
    endgenerate


    //have 64 rows of 64 bit partial products



    //add partial products
    integer k;

    //zero extend one partial product row to 128 bits
    //{64'd0, partial_product[k]}
    //shift left by k bits -> kth bit of b = left shift

    reg [127:0] product_sum;
    always @* begin
        product_sum = 128'd0;
        for (k=0; k<64;k=k+1)
            begin
                //shift by k bits to left before summing with total
                product_sum = product_sum + ({64'd0, partial_product[k]} << k);
            end
        end
    assign o = product_sum;

endmodule
```

IEEE Multiplier

```
//IEEE single precision floating point multiplier
//multiplies two 32-bit numbers by multiplying 23-bit mantissas
//and adding 8-bit exponents
//N = (-1)^S x 1.M x 2^(E-127)

//detrmine sign - XOR of signs
//zero, infinity, NaN


//define module
module iee_mul(
    input wire [31:0] a, //input
    input wire [31:0] b, //input
    output reg [31:0] o //output



    );


    wire sign_a = a[31]; //sign of a
    wire sign_b = b[31]; // sign of b - 1 bit
    wire [7:0] exp_a = a[30:23]; //exp a
    wire [7:0] exp_b = b[30:23]; //exponent b
    wire [22:0] man_a = a[22:0]; //mantissas
    wire [22:0] man_b = b[22:0];


    //sign product
    wire pro_sign = sign_a ^ sign_b; //xor
    // +  x + = +
        // + x - = -
        //- x - =+
```

```
    // +  x + = +
        // + x - = -
        //- x - =+


    //exponenttt
    // wire [7:0] exponent_s = exp_a +exp_b; //add exponents together
    // wire [7:0] final_exp  = exponent_s -8'd127; //E - 127


    //speical cases - notnormalized
    wire a_zero = (exp_a==8'd0) && (man_a==23'd0); //E and M = 0
    wire b_zero = (exp_b==8'd0) && (man_b==23'd0);
    wire a_inf = (exp_a==8'hFF) && (man_a==23'd0); //E = 255 (128) M = 0
    wire b_inf = (exp_b==8'hFF) && (man_b==23'd0); //infinity =N
    wire a_nan = (exp_a==8'hFF) && (man_a!=23'd0); //E = 255 M !=0
    wire b_nan = (exp_b==8'hFF) && (man_b!=23'd0); //NaN - not a numbers

    //denormalized??

    //spec flag and spec_final handle


    reg [31:0] spec_final;
    reg spec;
    always @(*)
        begin
// if either input is NAN - ouptutput is nan
            if (a_nan || b_nan)
                begin
                    spec_final = {1'b0, 8'hFF, 23'h400000}; // E= 255
                    spec = 1'b1;
                end
//infinity x 0 = undefined -- output = NAN
            else if ((a_inf && b_zero) || (b_inf && a_zero))
```

```
                    spec_final = {1'b0, 8'hFF, 23'h400000}; // E= 255
                    spec = 1'b1;
                end
//infinity x 0 = undefined -- output = NAN
            else if ((a_inf && b_zero) || (b_inf && a_zero))
                begin
                    spec_final = {1'b0, 8'hFF, 23'h400000}; //nan
                    spec = 1'b1;
                end
//inf x anything = infinity
            else if (a_inf || b_inf)
                begin
                    spec_final = {pro_sign, 8'hFF, 23'd0}; //infinity
                    spec = 1'b1;
                end
//0 x anything = 0
            else if (a_zero || b_zero)
                begin
                    spec_final={pro_sign, 8'd0, 23'd0}; //zero
                    spec = 1'b1;
                end
//normal case
            else
                begin
                    spec_final = 32'd0;
                    spec = 1'b0;
                end
    end


        //add leading 1 for normalized numbers -- 1.__
    wire [23:0] mant_a = (exp_a == 0) ? 24'd0 : {1'b1, man_a};
    wire [23:0] mant_b = (exp_b == 0) ? 24'd0 : {1'b1, man_b};
```

```
    //multply mantiassas
    wire [47:0] man_product = mant_a * mant_b;

    //nromalize
    wire msb = man_product[47];
    //exp calculation
    wire [8:0] exponent_sum = {1'b0, exp_a} + {1'b0, exp_b};
    wire [8:0] exp_n = exponent_sum - 9'd127 + {8'd0, msb}; //subtract , add 1 if normalized
//if msb = 1 -- takes bits [46:24] shift right by 1 1.xxxx
//msb = 0 -- result already normalized
    //get mantisa bsed on nromalization
    wire [22:0] man_norm = msb ? man_product[46:24] : man_product[45:23];


    ///overflow/underflow
    reg [7:0] exp_final;
    reg [22:0] man_final;

    always @(*)
        begin
            if(exp_n > 9'd255)
            begin
                //overlfow - retur ninfnity
                exp_final = 8'hFF;
                man_final = 23'd0;
            end
            else if (exp_n <= 9'd0)
            begin
            //underflow - return 0
                exp_final= 8'd0;
```

```
    ///overflow/underflow
    reg [7:0] exp_final;
    reg [22:0] man_final;

    always @(*)
        begin
            if(exp_n > 9'd255)
            begin
                //overlfow - retur ninfnity
                exp_final = 8'hFF;
                man_final = 23'd0;
            end
            else if (exp_n <= 9'd0)
            begin
            //underflow - return 0
                exp_final= 8'd0;
                man_final = 23'd0;
            end
            else
            begin
                exp_final = exp_n[7:0];
                man_final = man_norm;
            end
    end


    //final output
    //assign o = (spec_final != 0) ? spec_final : {pro_sign, exp_final, man_final};
    always @(*)
        begin
            if(spec)
                o = spec_final;
            else
                o = {pro_sign, exp_final, man_final};
```


```
                exp_final = exp_n[7:0];
                man_final = man_norm;
            end
    end


    //final output
    //assign o = (spec_final != 0) ? spec_final : {pro_sign, exp_final, man_final};
    always @(*)
        begin
            if(spec)
                o = spec_final;
            else
                o = {pro_sign, exp_final, man_final};
```


```
         end

endmodule
```